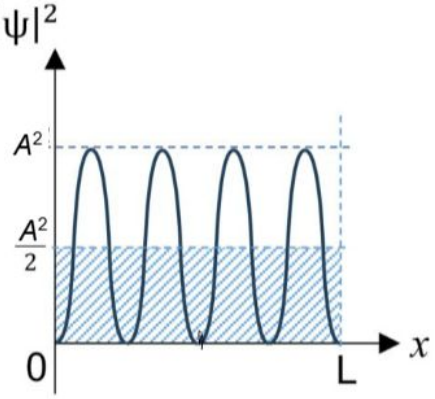
Qns	Answer	Marks
9(a)(i)	The square of the wavefunction amplitude $\lvert\psi(x)\rvert^2$ is the probability density function of the particle at position x.	B1
9(a)(ii)	The probability of the particle residing between x_1 and x_2 is given by the area under the $\lvert\psi(x)\rvert^2$-x graph between x_1 and x_2.	B1
9(b)(i)	$n = 4$.	B1

Qns	Answer	Marks
9(b)(ii)	 area under $= \frac{A^2}{2} L = 1$ $A = \sqrt{\frac{2}{L}}$	B1 M1 A1
9(b)(iii)	By symmetry of the graph, the area under the $\|\psi(x)\|^2$-x graph between $x = \frac{L}{8}$ and $x = \frac{L}{2}$ is $\frac{3}{8}$ of the total area, which is equal to 1. Hence, the probability is $\frac{3}{8} = 0.375$.	B1 B1
9(b)(iv)	$E = \frac{h^2}{8mL^2} n^2 = \frac{(6.63\times10^{-34})^2}{8\times9.11\times10^{-31}\times(3.5\times10^{-10})^2} \times 4^2 = 7.87778\times10^{-18}$ J $= \frac{7.87778\times10^{-18}}{1.60\times10^{-19}} = 49.2$ eV	C1 A1
9(c)(i)1	absorption spectrum dark lines on colourful/continuous background	B1 B1
9(c)(i)2	When white light containing all frequencies passes through the cold gas, some photons whose energies are exactly equal to the differences between the atomic energy levels will be absorbed. Since the energy levels are discrete, only photons of certain frequencies are absorbed. Photons of these frequencies will be re-radiated but in all directions when the atoms deexcite.	B1 B1 B1

Qns	Answer	Marks
	Consequently, the parts of the spectrum corresponding to these frequencies appear darker in comparison to the other frequencies.	B1
9(c)(ii)	emission spectrum colourful lines on dark background	B1
9(c)(iii)	$E_{photon} = -3.4 - (-13.6) = 10.2 \text{ eV} = 10.2 \times 1.6 \times 10^{-19} = 1.63 \times 10^{-18} \text{ J}$ $E_{photon} = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^{8}}{\lambda} = 1.63 \times 10^{-18} \text{ J}$ $\lambda = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^{8}}{1.63 \times 10^{-18}} = 1.22 \times 10^{-7} = 122 \text{ nm}$ in the ultraviolet region of the spectrum	M1 A1 A1